

main

July 26, 2025

1 Data Cleaning Pipeline

	User ID	Full Name	Date of Birth	Email	Age	\
0	12345	John Doe	2025-07-10	USER@GMAIL.COM	Twenty	
1	44556	alice SMITH	1995-03-25	alice.smith@yahoo.com	30	
2	12345	Bob Johnson	2000-11-15	invalid-email.com	45	
3	44556	Eve Adams	2024-01-01	eve@outlook.com	18	
4	77889	CHRIS EVANS	2010-09-30	chris_evans@EMAIL.com	Unknown	

	Subscription	Measurement	Phone Number
0	monthly	1.6bnp4hp	123-456-7890
1	yearly	2.5kw	9876543210
2	MOTHLY	3.7	(555) 123-4567
3	lifetime	1.5L	444-5555
4	mnthyl	4.3a8b	999.888.7777

1.1 Phase 1: Broad Cleaning (Applied to All Columns)

1.1.1 Functions

Standardize Column Names

- Convert to lowercase, remove spaces, replace special characters.

Fix Data Types #TODO!

- Convert numeric-looking text columns to numbers.
- Convert date columns to proper datetime format.
- Ensure boolean values are consistent.

Trim Whitespace & Standardize Case

- Remove leading/trailing spaces.
- Convert text to lowercase for consistency.

Correcting specific values

```
User ID
12345      2
44556      2
77889      1
Name: count, dtype: int64
```

```
Initial Mapping:
{'12345': '12345', '44556': '44556', '77889': '77889'}
```

```
User ID New User ID
0  12345          123
1  44556          445
2  12345          123
3  44556          445
4  77889        77889
```

1.1.2 Testing

	User ID	Full Name	Date of Birth	Email	Age \
0	12345	John Doe	2025-07-10	USER@GMAIL.COM	Twenty
1	67890	alice SMITH	1995-03-25	alice.smith@yahoo.com	30
2	11223	Bob Johnson	2000-11-15	invalid-email.com	45
3	44556	Eve Adams	2024-01-01	eve@outlook.com	18
4	77889	CHRIS EVANS	2010-09-30	chris_evans@EMAIL.com	Unknown

	Subscription	Measurement	Phone Number
0	monthly	1.6bnp4hp	123-456-7890
1	yearly	2.5kw	9876543210
2	MONTHLY	3.7	(555) 123-4567
3	lifetime	1.5L	444-5555
4	monthyl	4.3a8b	999.888.7777

	user_id	full_name	date_of_birth	email	age \
0	12345	John Doe	2025-07-10	USER@GMAIL.COM	Twenty
1	67890	alice SMITH	1995-03-25	alice.smith@yahoo.com	30
2	11223	Bob Johnson	2000-11-15	invalid-email.com	45
3	44556	Eve Adams	2024-01-01	eve@outlook.com	18
4	77889	CHRIS EVANS	2010-09-30	chris_evans@EMAIL.com	Unknown

	subscription	measurement	phone_number
0	monthly	1.6bnp4hp	123-456-7890
1	yearly	2.5kw	9876543210
2	MONTHLY	3.7	(555) 123-4567
3	lifetime	1.5L	444-5555
4	monthyl	4.3a8b	999.888.7777

```
user_id      12345
date_of_birth 2025-07-10
```

Name: 0, dtype: object

1.2 Phase 2: Targeted Cleaning (Specific to Data Type or Context)

	User ID	Full Name	Date of Birth	Email	Age	\
0	12345	John Doe	2025-07-10	USER@GMAIL.COM	Twenty	
1	67890	alice SMITH	1995-03-25	alice.smith@yahoo.com	30	
2	11223	Bob Johnson	2000-11-15	invalid-email.com	45	
3	44556	Eve Adams	2024-01-01	eve@outlook.com	18	
4	77889	CHRIS EVANS	2010-09-30	chris_evans@EMAIL.com	Unknown	

	Subscription	Measurement	Phone Number
0	monthly	1.6bnp4hp	123-456-7890
1	yearly	2.5kw	9876543210
2	MOTHLY	3.7	(555) 123-4567
3	lifetime	1.5L	444-5555
4	mnthyl	4.3a8b	999.888.7777

1.2.1 Functions

Dictionary for renaming columns

```
Index(['user_id', 'full_name', 'Date of Birth', 'Email', 'Age', 'subscription',  
      'Measurement', 'Phone Number'],  
      dtype='object')
```

Clean Categorical Variables

- Standardize known categories using a predefined dictionary.
- Correct spelling errors using predefined mapping and fuzzy matching for unknown errors.

Cleaned DataFrame:

	user_id	subscription
0	12345	Monthly
1	67890	Yearly
2	11223	Monthly
3	44556	Lifetime
4	77889	Monthly

Correction Log:

	original	corrected	score
0	Monthly	Monthly	100
1	Yearly	Yearly	100
2	Monthly	Monthly	100
3	Lifetime	Lifetime	100
4	mnthyl	Monthly	77

Clean Numeric Variables Extract Numeric Values from Measurement Columns

- Extract numeric values from mixed-format columns (e.g., "1.5kw" → 1.5).
- Flag non-standard measurement formats containing multiple non-numeric segments (e.g., "1.6bnp4hp"). How to handle these and substitute a value for them.

```
Measurement
0  1.6bnp4hp
1    2.5kw
2    3.7
3    1.5L
4    4.3a8b

Measurement cleaned_measurement
0  1.6bnp4hp      Invalid
1    2.5kw        2.5
2    3.7          3.7
3    1.5L         1.5
4    4.3a8b      Invalid

{'1.6bnp4hp': None, '4.3a8b': None}

Measurement cleaned_measurement
0  1.6bnp4hp        1.6
1    2.5kw        2.5
2    3.7          3.7
3    1.5L         1.5
4    4.3a8b        4.3
```

Apply Rules

- Example: Assign an “age_group” based on an age threshold. dd

```
Age
0  Twenty
1    30
2    45
3    18
4  Unknown

Age
0  20
1  30
2  45
3  18
4  9999

Age age_group
0  20      Adult
1  30      Adult
2  45      Old
```

3	18	Young Adult
4	9999	Unknown

1.3 Phase 3: Final Checks & Quality Assurance

8. Re-check Missing Values
 - Identify remaining gaps and decide on appropriate handling (e.g., imputation, removal).
9. Summary Statistics
 - Generate descriptive stats to validate the dataset's integrity.